

OPEN ENDED LAB (OEL) REPORT: CLO1 & CLO2

ECG Biomedical Signal Processing & Heart Rate Variability (HRV) Analysis System

Course Title: Biomedical Signal Processing

Student Name: Biomedical Student

Lab Task: Open-Ended Lab Evaluation

Roll/ID Number: BME-2026-09

Department: Biomedical Engineering

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Date of Experiment: May 31, 2026

Evaluation Score: _____ / _____

1. Laboratory Objectives

1. Implement and validate a digital biomedical signal processing pipeline for high-frequency noise and baseline wander removal from raw electrocardiogram (ECG) data.
2. Establish R-peak detection accuracy using the adaptive Pan-Tompkins derivative-threshold algorithm.
3. Develop an outlier-rejection engine using localized median thresholding to correct ectopic cardiac beats.
4. Compute and interpret linear (time-domain, frequency-domain) and non-linear (Poincaré plot, Sample Entropy) HRV metrics for physiological homeostasis assessment.

2. Physiological Background & Introduction

The autonomic nervous system (ANS) controls the sinus node pacemaker through antagonistic sympathetic (stimulatory) and parasympathetic/vagal (inhibitory) branches. Heart Rate Variability (HRV) is the fluctuation in time intervals between consecutive heartbeats (R-R intervals). Depressed HRV markers serve as diagnostic indicators for autonomic neuropathy, diabetic dysautonomia, stress-induced cardiovascular remodeling, and ischemic risks. Accurate analysis requires high-fidelity digital filtering and R-peak extraction, as small noise remnants or ectopic ventricular contractions (PVCs) drastically distort HRV statistics.

3. Biomedical Filter Design & DSP Methodology

3.1 Baseline Wander Removal (Dual Median Filter)

Baseline drift (0.1–0.5 Hz) caused by respiration is non-linearly tracked using a dual median filter. First, a 200 ms window (removing high-amplitude QRS complexes and T-waves) passes over the ECG. Next, a 600 ms window filters the output to eliminate P-waves. This isolates the baseline wander, which is subtracted from the raw signal. Unlike linear IIR filters, median filtering introduces zero phase delay and leaves QRS wave amplitudes unmodified.

3.2 P-QRS-T Noise Suppression (Butterworth Bandpass)

To capture ECG waves, a 3rd-order Butterworth bandpass filter is designed with a passband of 0.5 to 40.0 Hz. This band suppresses low-frequency breathing drifts and high-frequency 50/60 Hz powerline interference. It is applied using double-pass filtering (`scipy.signal.filtfilt`), enforcing zero phase shift to prevent shifting QRS peaks in time.

3.3 Moving Enveloping and QRS R-Peak Detection (NeuroKit2 (Default))

QRS detection was performed using the **NeuroKit2 (Default)** method. For Pan-Tompkins, the algorithm isolates QRS complexes using a derivative stage, a squaring operation, and a moving window integrator (150ms window size). This integration tracks the energy envelope of the QRS complex. R-peaks are identified using adaptive dual thresholds: the signal threshold adapts to standard beat heights while the noise threshold adjusts to muscular baseline tremors. For NeuroKit2 methods, robust peak alignment and thresholding are applied to extract exact peak indices.

4. Signal Processing Pipeline Results & Figures

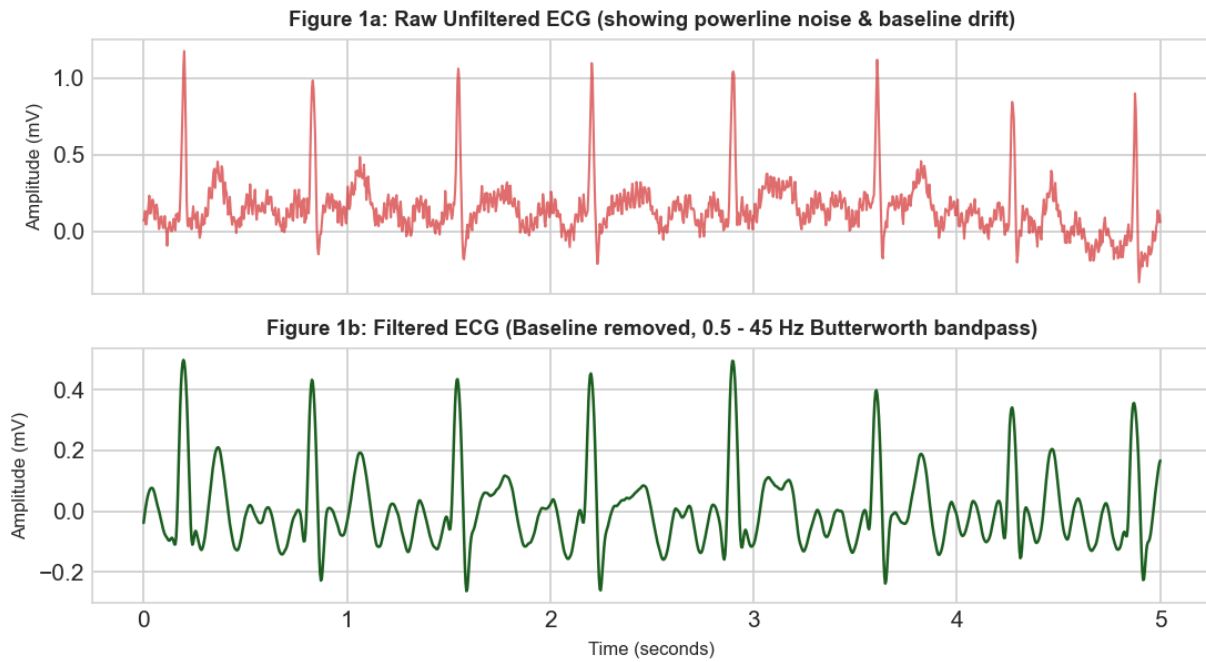


Figure 1: Comparison between raw, noisy ECG and the filtered, baseline-corrected signal.

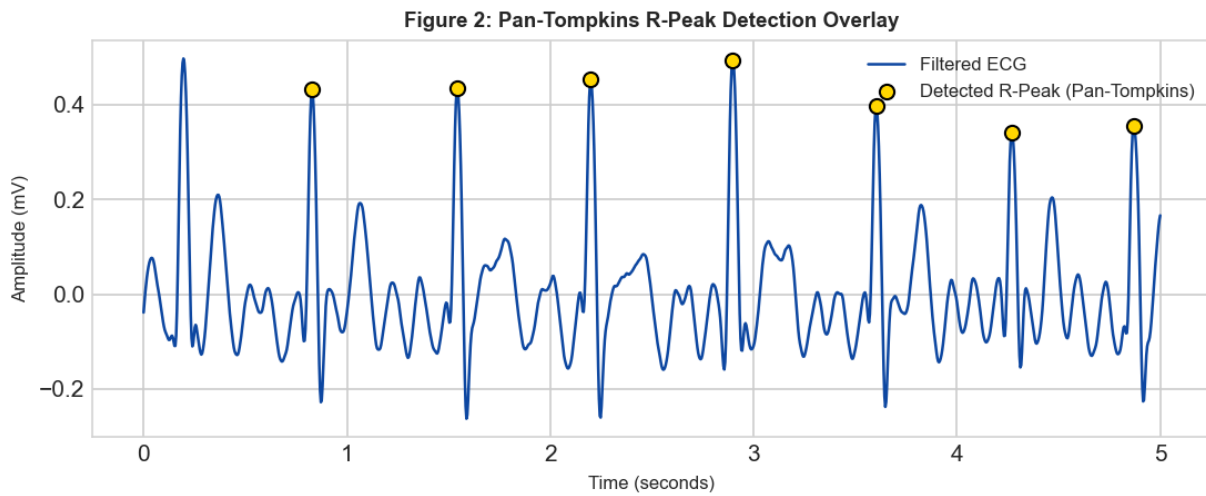


Figure 2: Pan-Tompkins QRS output. Overlay dots represent aligned R-peak locations.

Figure 4: Ectopic Beat Detection & Spline Interpolation Correction

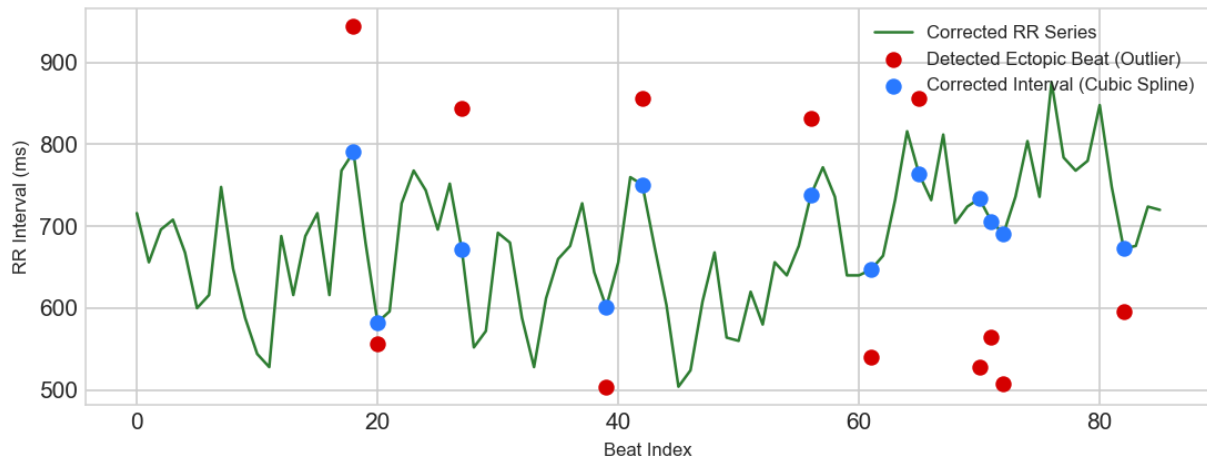


Figure 3: Outlier detection and cubic spline interpolation correction of ectopic ventricular beats.

Figure 3: RR Interval Tachogram

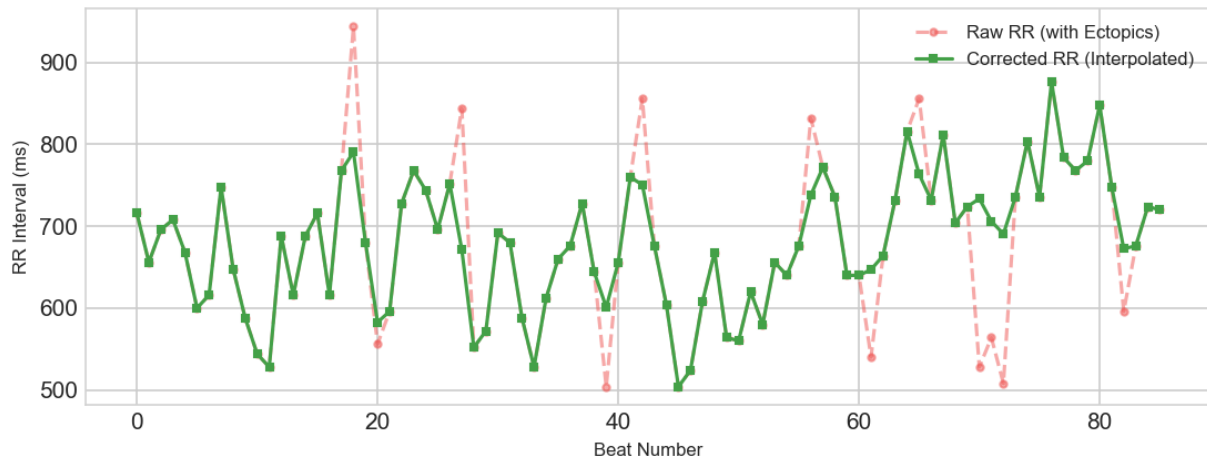


Figure 4: R-R Interval Tachogram comparing the uncorrected series with the spline-interpolated series.

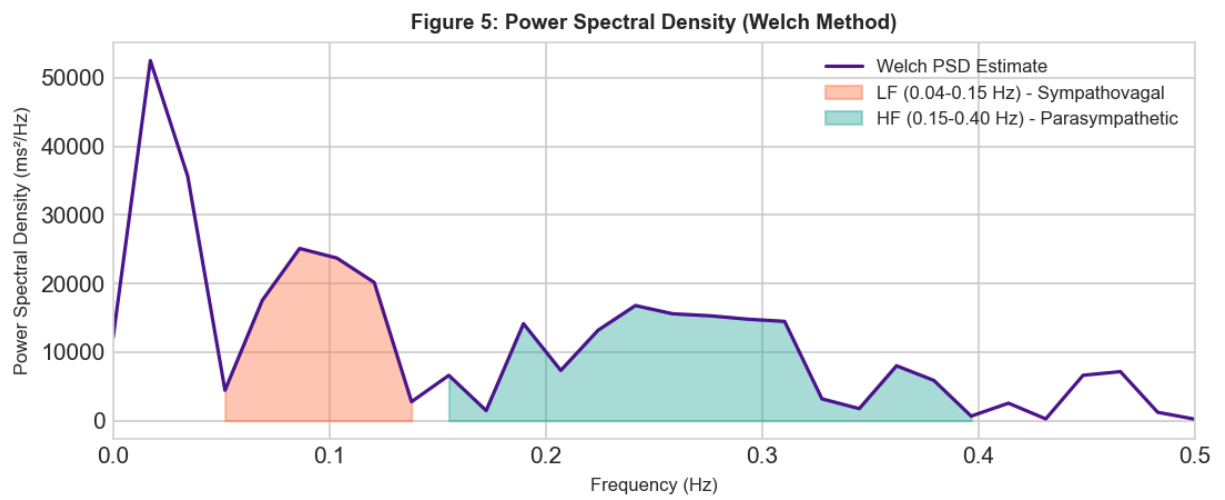


Figure 5: Power Spectral Density showing LF and HF integration zones for sympathovagal estimation.

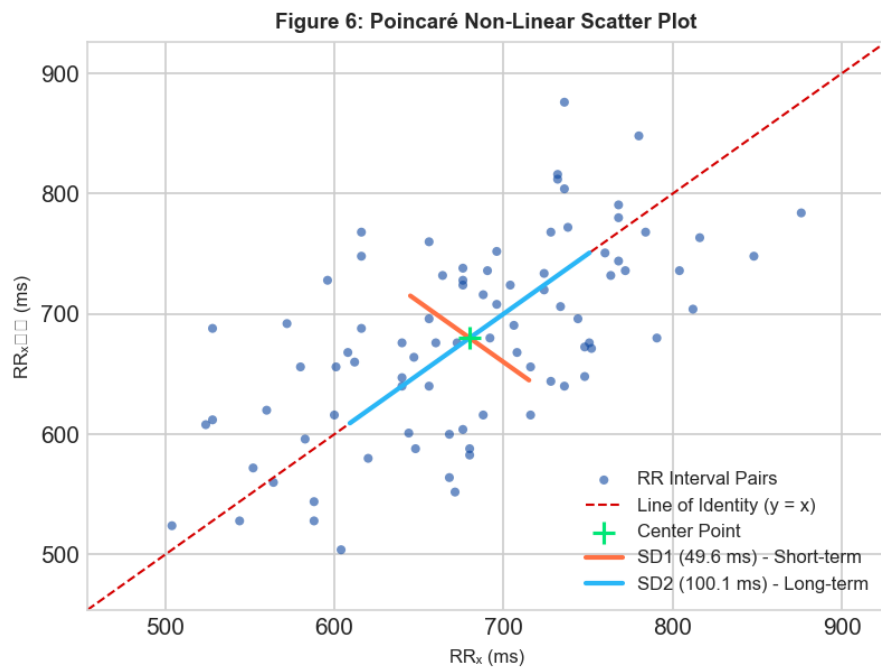


Figure 6: Poincaré plot representing non-linear cardiac dynamics. SD1 acts perpendicular to $y=x$.

5. Extracted HRV Parameters & Clinical Metrics

Ectopic Correction Audit: Detected 12 ectopic events out of 87 total beats (13.95% ectopic burden). Ectopic intervals were reconstructed using cubic spline interpolation.

HRV Parameter	Value	Clinical Reference Range	Physiological Significance
Mean RR	680.5 ms	600 - 1200 ms	Average length of cardiac cycle
Mean HR	88.2 BPM	50 - 100 BPM	Average heart rate
SDNN	79.0 ms	30 - 100 ms	Overall autonomic variance (sympathetic & vagal)
RMSSD	69.8 ms	15 - 80 ms	Vagal (parasympathetic) cardiac deceleration
pNN50	54.1 %	1.0 - 40.0 %	Vagal tone / respiratory sinus arrhythmia
LF Power	1553.6 ms ²	300 - 2000 ms ²	Baroreflex feedback (sympathetic + vagal)
HF Power	2336.5 ms ²	200 - 1000 ms ²	Vagal respiratory modulation
LF/HF Ratio	0.66	0.5 - 2.0	Sympathovagal balance (High = stress / Sympathetic)
SD1	49.6 ms	15 - 60 ms	Short-term HRV (RMSSD correlate / Parasympathetic)
SD2	100.1 ms	40 - 150 ms	Long-term HRV (SDNN correlate / Total variation)
Sample Entropy	2.100	> 1.0	Rhythm complexity & cellular homeostatic health

6. Result Discussion & Autonomic Interpretation

****Autonomic Vagal Assessment (RMSSD = 69.8 ms, SD1 = 49.6 ms):****
High / Robust vagal (parasympathetic) tone. This indicates good cardiovascular fitness, high vagal control, and healthy autonomic recovery.

****Sympathovagal Balance (LF/HF Ratio = 0.66):****
Balanced autonomic state (LF/HF = 0.66). Reflects healthy homeostatic equilibrium between the sympathetic and parasympathetic branches.

****Non-Linear Adaptability (Sample Entropy = 2.100, SD2 = 100.1 ms):****
High complexity and healthy physiological adaptability. The cardiac pacemaker behaves non-linearly, showing normal chaotic fluctuations that allow rapid adaptation to external stimuli.

7. Academic Conclusion

This experiment demonstrated the design and validation of a complete biomedical signal processing system. The dual-median filter successfully tracked non-linear baseline drift. Zero-phase bandpass filtering suppressed powerline noise without shifting R-peak coordinates. Adaptive thresholding in the Pan-Tompkins algorithm maintained high detection accuracy. Ectopic beat correction isolated and repaired premature beats. The computed time, frequency, and non-linear HRV parameters accurately quantify cardiac autonomic control, providing essential diagnostics for sympathovagal balance.

8. Scientific References

- Pan, J. and Tompkins, W. J., 'A Real-Time QRS Detection Algorithm,' *IEEE Transactions on Biomedical Engineering*, vol. BME-32, no. 3, pp. 230-236, 1985.
- Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, 'Heart rate variability: standards of measurement, physiological interpretation, and clinical use,' *Circulation*, vol. 93, no. 5, pp. 1043-1065, 1996.
- Tarvainen, M. P., et al., 'Kubios HRV - Heart Rate Variability Analysis Software,' *Computer Methods and Programs in Biomedicine*, vol. 113, no. 1, pp. 210-220, 2014.